

IDLAN ZAFRAN MOHD ZAIDIE

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Portfolio: bit.ly/idlanzafran | **GitHub:** github.com/IdlanZafran | **LinkedIn:** linkedin.com/in/idlanzafran/

Mechanical Engineering undergraduate (3.87 CGPA) bridging advanced mechanical design with enterprise-grade telemetry and Machine Learning. Proven expertise in full-lifecycle engineering, from conducting FEA stress simulations on complex mechanical systems to writing memory-safe C++ firmware for robust data acquisition and edge computing. Seeking an internship to apply cross-functional skills in predictive maintenance, equipment automation, and real-time asset monitoring.

EDUCATION

UNIVERSITI KEBANGSAAN MALAYSIA (UKM)

Expected Graduation: 2027

Bachelor of Mechanical Engineering | CGPA: 3.87/4.00 (Current GPA: 3.90/4.00)

TECHNICAL SKILLS

- **Mechanical & FEA:** 3DEXPERIENCE CATIA, Autodesk Inventor, Fusion 360, AutoCAD, Solid Mechanics, Rotating Equipment.
- **Firmware & Edge Computing:** C/C++, PlatformIO, ESP8266/ESP32, LittleFS, Local HTML/JS Web Servers.
- **Data Science (Predictive Maintenance):** R, Python, Machine Learning (k-NN, K-Means Clustering, Regression).
- **Prototyping:** HSE Sensor Integration (Gas/Smoke, Proximity, Motion, DHT, RMS Power), Circuit Design, Rapid PCB Modification.

PROFESSIONAL EXPERIENCE

ThingsSentral | UKM

July 2025 - Present

Research Assistant & Embedded Systems Developer (Sept 2025 - Present) | Hardware & Firmware Intern (July - Sept 2025)

- **End-to-End Hardware R&D:** Spearheaded the development of custom IoT sensor nodes from scratch, encompassing 3D CAD enclosure design, circuit building, and full-stack firmware programming.
- **IoT Fleet Deployment & SCADA:** Engineered an IoT SCADA architecture, deploying 8 custom-built remote nodes (DHT, smoke, light, proximity, motion, door, alarms, smart plugs) to monitor critical infrastructure, including university server and backup battery rooms. Established a scalable framework for emergency threshold monitoring applicable to offshore environments, achieving 2-second telemetry via HTTP keep-alive.
- **Fail-Safe Edge Computing:** Developed an offline LittleFS data buffering vault, ensuring continuous bulk data logging during network outages and extending backup life to over a week.
- **Enterprise Telemetry Libraries:** Authored memory-safe C++ telemetry libraries (ThingsSentral, RapidBootWiFi) actively utilized by university researchers for remote asset data transmission.
- **Access Control Integration:** Designed a 12V RFID smart lock with a Telegram-integrated dashboard, significantly optimizing power consumption through a 2-minute Wi-Fi sync cycle.
- **Bi-Directional Telemetry & Actuation:** Engineered a remote command-execution node integrated with the ThingsSentral dashboard, demonstrating two-way SCADA capabilities for remote asset control and wireless actuation.

ENGINEERING PROJECTS

Full-Lifecycle Mechatronics: Autonomous Parcel Sorting Robot | Academic Project

Sept 2025 - Present

- **Project Leadership:** Leading a 5-person team to design, manufacture, and program an autonomous parcel sorting system while strictly managing a sub-RM750 prototyping budget.
- **End-to-End Fabrication:** Executed a comprehensive 3-month mechanical design and FEA phase (Autodesk Inventor) during Semester 5. Currently managing the physical fabrication and assembly utilizing Aluminum Alloy hollow framing in Semester 6.

Mechanical Power Transmission & FEA: Double Helical Elevator Gearbox | Academic Project

May 2025

- **Design & Performance:** Engineered a double helical transmission system in Autodesk Inventor within a 3-weeks. Optimized gear ratios for 98.54% efficiency and a 1.733 meters per second lifting speed under a 50kg load.
- **FEA & Structural Validation:** Validated structural integrity via FEA, verifying zero yielding with a safety factor of 4 and a max Von Mises stress of 10.08 MPa.

Predictive Maintenance: Machine Learning Analysis on Material Stress | Academic Project

May 2025

- **Predictive Maintenance Modelling:** Developed an R-based linear regression model (R-squared of 0.967) on stress-microstrain data to rapidly predict material fatigue for pipeline and heavy machinery asset integrity.
- **Risk Classification:** Applied k-Nearest Neighbors to accurately classify stress levels into risk categories, validating stress-strain zones via K-Means clustering supported by the Elbow Method.

Mechanical Design & FEA: Hydro-Powered River Cleanup Conveyor | Academic Project

Nov 2024 - Jan 2025

- **Mechanical Design & Force Analysis:** Engineered a conceptual floating, river-current-powered turbine and conveyor system, performing manual static and hydrodynamic calculations to determine critical buoyancy, drag, and mechanical tension.
- **Autodesk Inventor & AutoCAD:** Conducted Von Mises stress simulations in Inventor to validate structural integrity (specifying Aluminium Alloy 6061) and generated 2D technical documentation using AutoCAD.

LEADERSHIP & GLOBAL COLLABORATION

International Project Lead | ICAPS (17-University Consortium)

Jan 2026 - Present

- **Global Collaboration & Proposal:** Spearheading a multidisciplinary, multinational team (including South Korean counterparts) to conceptualize and pitch an AI-powered manufacturing unit for an ASEAN-wide competition.
- **Mechatronic & AI System Design:** Currently architecting a system utilizing Spatial AI for 3D volume mapping and a PID-controlled thermal press for biodegradable packaging.
- **Project Management:** Managing a USD 700 prototyping budget while coordinating cross-border CAD development and the ongoing physical fabrication of the system.

President | UKM Robocon Club

Jan 2025 - June 2025

- **Team Leadership:** Directed mechanical strategies and technical development for a 36-member team, successfully deploying an autonomous robot for the ROBOCON Malaysia 2025 national stage.

ADDITIONAL AWARDS & HIGHLIGHTS

- **Invited Keynote Speaker:** AISCOST Indonesia (Sept 2025). Served as the sole Malaysian speaker on “Youth Innovation and Research”.
- **Technical Trainer:** Institut Teknologi Bandung (ITB) (Dec 2025). Led hands-on IoT and SCADA workshops for undergraduate/PhD cohorts.
- **Academic Honors:** Dean’s List Award (All Semesters).
- **Competition Awards:** Top 10 Finalists China-ASEAN Innovation Quest (2025), Gold Medal (6th Sustainability Challenge 2025), 1st Place (MIKEF Gear Up Challenge 2025), 2nd Place (MIKEF Machine Learning Data Explore Challenge 2025), Gold Medal Green City AR Challenge (2024).
- **Certifications:** ThingsSentral Internship Program Competency Certificate (Level 3).

REFEREES

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